

Exponential Growth and Decay Practice

Evaluating exponential functions, graphing them from a table of values, and using growth and decay models.

- An exponential function has the form $f(x) = a \cdot b^x$: the *starting value* a is multiplied by the *base* b once for every step of x . Raise the base to the power *first*, then multiply by a .
- A negative exponent means divide: $b^{-n} = \frac{1}{b^n}$.
- Show the substitution and the power before you write the final value.
- Where a grid is given, plot your points and join them with a smooth curve — exponential graphs are never straight lines.
- Round money answers to the nearest cent and people or animals to a whole number, as each problem says.

1. Let $f(x) = 3 \cdot 2^x$. Find $f(4)$.

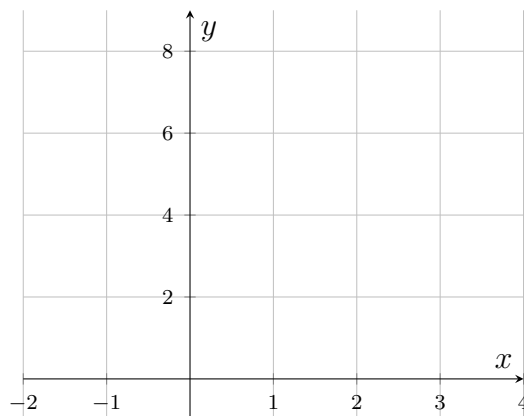
Answer: _____

2. A function is given by $g(x) = 100 \cdot (0.8)^x$. Find $g(2)$.

Answer: _____

- 3.** Complete the table of values for $h(x) = 2^x$, then plot the five points on the grid and join them with a smooth curve.

x	-1	0	1	2	3
$h(x)$					



Answer: _____

- 4.** Let $p(x) = 2 \cdot 3^x$. Find $p(-2)$. Give the exact answer as a fraction.

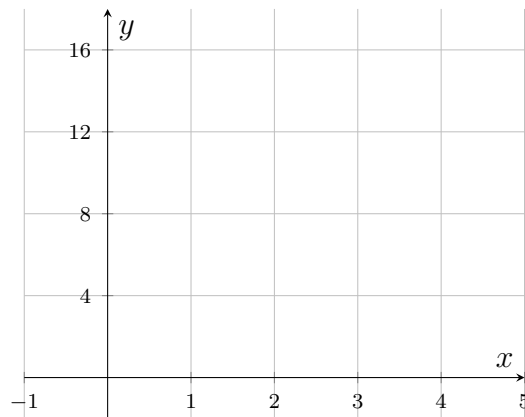
Answer: _____

5. A biology class starts a culture with 400 bacteria. The number of bacteria doubles every hour, so after t hours the count is $P(t) = 400 \cdot 2^t$. How many bacteria are there after 5 hours?

Answer: _____

6. A tank holds 16 litres of water and half of what remains drains away each minute, so after x minutes the amount left is $k(x) = 16 \cdot \left(\frac{1}{2}\right)^x$. Build a table for $x = 0, 1, 2, 3, 4$, graph the function on the grid, and then use your graph to find the minute at which exactly 2 litres are left.

x	0	1	2	3	4
$k(x)$					



Answer: _____

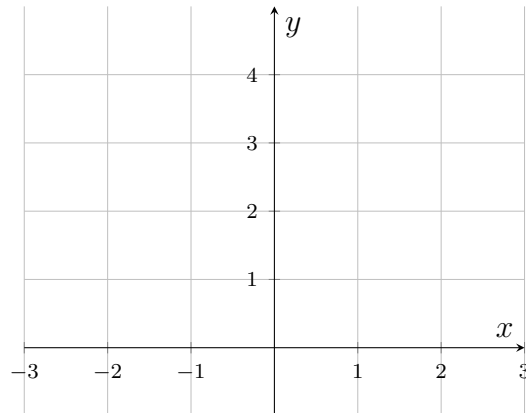
7. A car bought for 24000 dollars keeps 0.85 of its value each year, so its value after t years is $V(t) = 24000 \cdot (0.85)^t$. Find the value of the car after 4 years, in dollars, rounded to the nearest cent.

Answer: _____

8. Let $f(x) = 6 \cdot (0.5)^x$. Find $f(-1)$ and $f(3)$. Give $f(3)$ as an exact fraction.

Answer: _____

- 9.** On the grid, sketch both $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$ for x from -2 to 2 . Then use the two curves to find the one value of x where the two functions are equal.

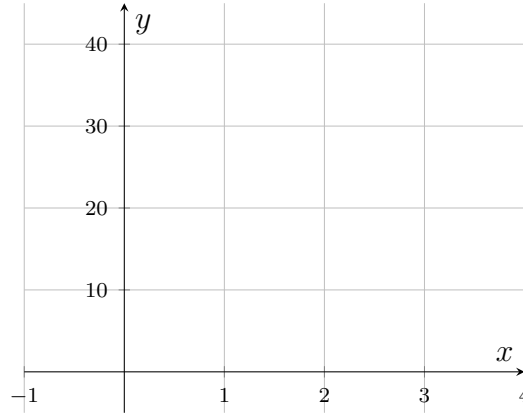


Answer: _____

- 10.** A town of 8500 people grows by 3 percent every year, so its population after t years is $8500 \cdot (1.03)^t$. Find the population predicted after 12 years, rounded to the nearest whole person.

Answer: _____

- 11.** The graph of $y = a \cdot 2^x$ passes through the point $(3, 40)$. Find the starting value a , then sketch the completed function on the grid.



Answer: _____

- 12.** Two accounts are opened on the same day. Account A starts with 1200 dollars and grows by 5 percent each year, so it is worth $1200 \cdot (1.05)^t$ after t years. Account B starts with 900 dollars and grows by 9 percent each year, so it is worth $900 \cdot (1.09)^t$. Find the value of each account after 10 years, in dollars, rounded to the nearest cent, and state which account is worth more.

Answer: _____